

# IP Addressing & Subnetting

## Task 1: IP Address Classes

| Class | IP Address Range            | Default Subnet Mask | Example Device/Service            |
|-------|-----------------------------|---------------------|-----------------------------------|
| A     | 1.0.0.0 – 126.255.255.255   | 255.0.0.0           | Large ISP network server          |
| B     | 128.0.0.0 – 191.255.255.255 | 255.255.0.0         | University network                |
| C     | 192.0.0.0 – 223.255.255.255 | 255.255.255.0       | Home Wi-Fi router                 |
| D     | 224.0.0.0 – 239.255.255.255 | Not used for hosts  | Video streaming multicast service |
| E     | 240.0.0.0 – 255.255.255.255 | Reserved            | Research and testing purposes     |

## Task 2: Calculate Network Address, Broadcast Address, and Host Range

Given:

IP Address = 192.168.10.45

Subnet Mask = 255.255.255.0 (/24)

Step 1: Network Address = 192.168.10.0

Step 2: Broadcast Address = 192.168.10.255

Step 3: Valid Host Range = 192.168.10.1 to 192.168.10.254

## Task 3: Subnetting Plan for Food Delivery Startup

Given Network: 10.0.0.0/24

Subnet Mask: 255.255.255.192 (/26)

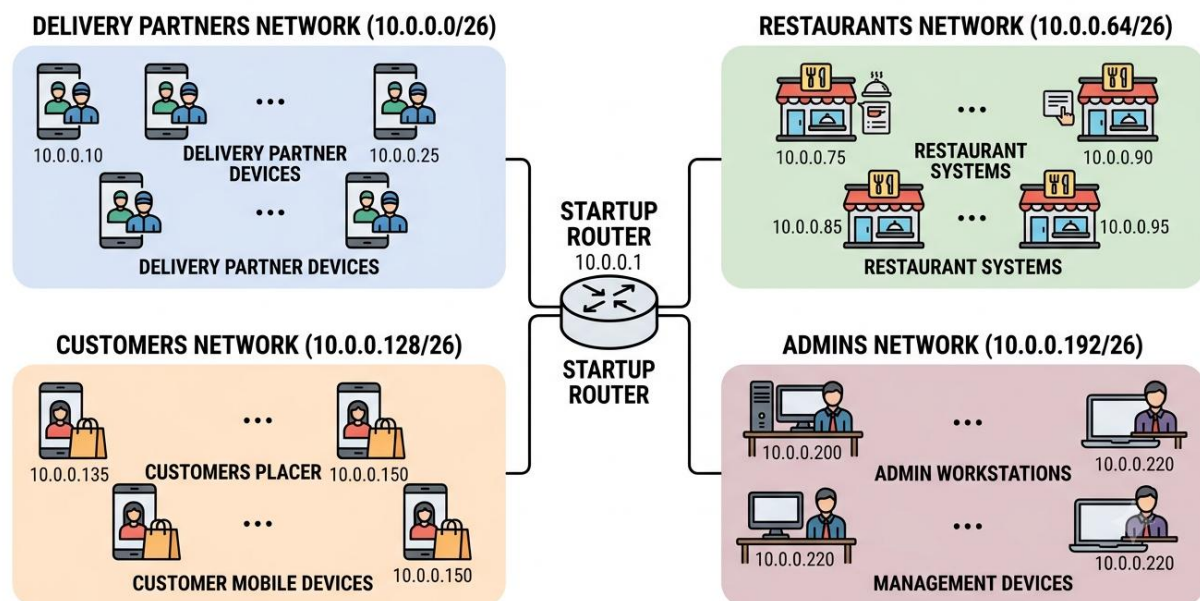
Delivery Partners: 10.0.0.0/26

Restaurants: 10.0.0.64/26

Customers: 10.0.0.128/26

Admins: 10.0.0.192/26

### FOOD DELIVERY STARTUP NETWORK ARCHITECTURE - SUBNET DIAGRAM



## Task 4: College WiFi Network Subnetting

Network = 172.16.0.0/16

Need at least 20 subnets

$2^4 = 16$  (not enough)

$2^5 = 32$  (enough)

Borrowed Bits = 5

New Prefix = /21

New Subnet Mask = 255.255.248.0

Hosts per Subnet =  $2^{(32-21)} - 2 = 2046$

## Task 5: Invalid IP Address Explanation

192.168.1.300 is invalid because each octet must be between 0 and 255.

300 is greater than 255.

Valid examples:

192.168.1.10

192.168.1.50

192.168.1.100

192.168.1.200

## Conclusion:

In this assignment, I studied IP address classes, subnetting, network and broadcast addresses, host ranges, and subnet planning. I also learned how to identify invalid IP addresses and calculate subnet masks for different network requirements.